

Fangyi Yu

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Add.: 545 Vanderbilt Avenue, Brooklyn, New York

Education

New York University

2025.9~Present

M.S. in Electrical Engineering GPA: 3.556/4.0

- Core Courses: Digital Signal Processing, Computer Systems Architecture, Introduction to Machine Learning, Design and Analysis of RFIC, LLM4ChipDesign, Applied Matrix Theory

Lanzhou University

2021.9~2025.6

B.S. in Microelectronics Science and Engineering GPA: 84.96/100

- Core Courses: Quantum Mechanics (94), Method of Mathematical Physics (97), Solid-State Electronic Device (93), Physics of Semiconductors (89), Microelectronics Manufacturing Technology (87), Engineering Electromagnetic Theory (85), Analog Circuit (86)

Research Experiences

AI-based Hardware Attacks (AHA!) Challenge

Course Project New York University

2026.4

Designed and evaluated LLM-driven hardware Trojans in a red-team setting to advance understanding of AI-enabled hardware security vulnerabilities and attack methodologies.

- Conducted a vulnerability assessment of the Secworks AES core by integrating LLM-generated hardware Trojans, optimizing its evasive stealth and minimal hardware footprint to bypass RTL-level detection
- Trojan 1: Designed a functional-logic-based hardware Trojan targeting the AES core's control path. Implemented it as a dormant circuit that remains inactive during normal cryptographic operations to maintain a low power and area profile
- Trojan 2: Implemented a stealthy "Time Bomb" hardware Trojan using an 8-bit counter that triggers after 100 valid encryption cycles (8'h64), at which point it hijacks the AES output to leak the lower 32 bits of the secret key while leaving the remaining bits intact to evade detection
- Trojan 3: Developed a high-stealth Denial-of-Service (DoS) hardware Trojan utilizing fragmented detection logic to identify a specific three-stage sequential input pattern, which once matched, permanently hijacks the internal control FSM to lock the core in a busy state (ready = 0) until a hardware reset.

2.4-GHz Direct-Conversion CMOS Receiver Signal-Path Design

Course Project, New York University

2026.4~2026.5

Architected a full-chain RF front-end (LNA, Gilbert-cell Mixer, IF Stage) under stringent 100 nH total on-chip inductance and 1.8V supply constraints.

- Designed and simulated the RF-to-baseband signal path of a 2.4-GHz direct-conversion CMOS receiver, integrating an inductively degenerated cascode LNA, double-balanced Gilbert-cell mixer, and differential IF gain stage
- Optimized LNA input matching and gain through transistor sizing, bias tuning, and inductor/capacitor sweeps, achieving low input return loss, high forward gain, and reducing standalone noise figure across the 2.40–2.48 GHz band
- Built and verified the complete receiver signal-path schematic in Cadence Virtuoso, using SpectreRF simulations for DC biasing, S-parameter matching, noise figure, conversion gain, bandwidth, and two-tone IIP3 characterization
- Balanced gain, noise, linearity, impedance matching, and passive-component constraints, achieving 37.1 dB receiver gain, 50 Ω input matching, 700 Ω output resistance, and compliance with the 100 nH total on-chip inductance limit

Research on the Sensor Performance Based on WS₂

Supervisor: Prof. Yanrong Wang, Lanzhou University

2025.3~2025.6

WS₂-based gas sensors with high sensitivity to ethanol for monitoring ethanol vapor in public environments, contributing to the prevention of fire hazards and gas poisoning.

- Preparing high-purity tungsten disulfide (WS₂) using a controlled hydrothermal synthesis process for the

fabrication and optimization of a high-performance gas sensor

- Characterizing the crystal structure and morphology of WS₂ material using scanning electron microscopy (SEM) and transmission electron microscopy (TEM)
- Conducting performance testing for the fabricated gas sensor including testing its optimal operating temperature, selectivity for different gases, and gradient performance for ethanol gas
- Analyzing sensor response characteristics under varying operating conditions, including response curves, sensitivity, and stability with respect to temperature and gas concentration.

Electronic Circuit and Structure Designing

Supervisor: Prof. Tianyi Zhang, Institute of Automation, Chinese Academy of Sciences

2024.7~2024.8

Conducted human hand kinematic simulations to investigate the fundamental principles of dexterous hand manipulation and robotic hand design.

- Reviewed global dexterous hands research via academic literature and company official website, organized findings into a slide report, establishing the field's knowledge foundation
- Built a knowledge framework of humanoid robots by studying the status of humanoid robots both domestic and abroad and refining the core points, presented the knowledge framework in the form of a slide report
- Wrote a review on dexterous hands, including the current development status, key technologies, mechanical structures, control methods, and future challenges and trends of dexterous hands both domestically and internationally
- Validated the functionality and grasping capability of a human hand kinematic model by simulating basic palm motions, thumb opposition, and multiple grasping taxonomies

Design of 4-bit Absolute Value Detector | online

Supervisor: Prof. Dejan Marković, University of California, Los Angeles

2024.1~2024.3

- participated in the circuit design, mainly constructed the circuit structure framework, and completed the corresponding description part
- Adapted adders, comparators, and multiplexers to build the circuit and designed the basic circuit topology to achieve a circuit delay of 1.5 times the minimum value
- Improved the performance and efficiency of the circuit by optimizing the integration of gate size and supply voltage

Summer Engineering Project, Cambridge University

2023.7~2023.8

- Participated in an interdisciplinary engineering program covering jet engines, nanotechnology, and quantum physics, with a focus on developing fundamental research skills
- Strengthened ability in literature review, technical reporting, and scientific communication through project-based learning

Publication

- **Fangyi Yu**, Research on the Development of LDO Chip Technology, 2024, The 2nd International Conference on Machine Learning and Automation.
- **Fangyi Yu**, Research and Analysis on Artificial Intelligence in Integrated Circuits, 2024, 2nd International Conference on Artificial Intelligence, Database and Machine Learning (AIDML 2024).

Skills

- **Language:** Mandarin (Native), English (Fluent IELTS: 7.0)
- **Programming:** Python, Pytorch, Verilog
- **Software:** MATLAB, Cadence, Origin